1. Write a program to check whether a given number is positive or negative.

```
num = int(input("Enter an Integer : "))
if num < 0:
  print("Negative!")
else:
  print("Positive!")
```

```
Enter an Integer : -30
Negative!
```

2. Write a program to check whether a number is even or odd.

```
num = int(input("Enter a number : "))
if num % 2 == 0:
  print("Even Number!")
else:
  print("Odd Number!")
```

```
Enter a number : 3
Odd Number!
```

3. Write a program to check whether a person is eligible to vote or not. Take age as input.

```
age = int(input("Enter your age : "))
if age >= 18:
  print("Eligible to vote!")
else:
  print("Not eligible to vote!")
```

```
Enter your age : 30
Eligible to vote!
```

4. Write a program to find the greater number between two numbers entered by the user.

```
num1 = int(input("Enter 1st number : "))
num2 = int(input("Enter 2nd number : "))
if num1 > num2:
  print("1st num is greater!")
elif (num2 > num1):
  print("2nd num is greater!")
else:
  print("Both numbers are equal!")
```

```
Enter 1st number : 10
Enter 2nd number : 20
2nd num is greater!
```

5. Write a program to check whether a student has passed or failed. Passing marks are 40.

```
marks = int(input("Enter your marks : "))
if marks < 40:
  print("Fail!")
else:
  print("Pass!")
```

```
Enter your marks : 50
Pass!
```

6. Write a program to check whether a number is divisible by both 5 and 10.

```
num = int(input("Enter an number : "))
if num % 5 == 0 and num % 10 == 0:
  print("num divisible 5 and 10!")
else:
  print("not divisible by 5 and 10!")
```

```
Enter an number : 50
num divisible 5 and 10!
```

7. Write a program to check whether a given character is a vowel or a consonant.

```
char = input("Enter a character : ").lower()
if char in ['a','e','i','o','u']:
  print("Vowel!")
else:
  print("Consonant!")
```

```
Enter a character : a
Vowel!
```

8. Write a program to calculate electricity bill based on units consumed.

```
units = int(input("Enter units : "))
if units < 100:
  bill = units * 2
else:
  bill = units * 5
print("Bill : ", bill, " Rs")
```

```
Enter units : 200
Bill :  1000  Rs
```

9. Write a program to check whether a year is a leap year or not.

```
year = int(input("Enter year : "))
if (year % 4 == 0) or (year % 100 != 0) or (year % 400 == 0):
  print("Leap Year!")
else:
  print("Not lear year!")
```

```
Enter year : 2000
Leap Year!
```

10. Write a program to calculate bonus for an employee based on salary.

```
sal = float(input("Enter salary : "))
if (sal < 50000) :
  bonus = sal * 0.10
else:
  bonus = sal * 0.05
print("Bonus : ", bonus)
```

```
Enter salary : 40000
Bonus :  4000.0
```

11. Write a program to print numbers from 1 to 10 using a loop.

```
for i in range(1,11):
  print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

12. Write a program to print all even numbers between 1 and 50.

```
for i in range(1,51):
  if i % 2 == 0:
    print(i)
```

```
2
4
6
8
10
12
14
16
18
20
22
24
```

```
26
28
30
32
34
36
38
40
42
44
46
48
50
```

13. Write a program to calculate the sum of first 10 natural numbers.

```
sum = 0
for i in range(1,11):
  sum = sum + i
print(sum)
```

```
55
```

14. Write a program to print the multiplication table of a number entered by the user.

```
num = int(input("Enter a number : "))
for i in range(1,11):
  print(num*i)
```

```
Enter a number : 5
5
10
15
20
25
30
35
40
45
50
```

15. Write a program to count the number of digits in a given number.

```
num = input("Enter a number : ")
count = 0
for digit in num:
  count = count+1
print("Count of digits : ", count)
```

```
Enter a number : 100
Count of digits :  3
```

16. Write a program to check pass or fail for 5 students using loop and condition.

```
for i in range(1,6):
  marks = int(input(f"Enter marks for student {i} : "))
  if marks < 40:
    print("Student ", i, " Fail!")
  else:
    print("Student ", i, " Pass!")
```

```
Enter marks for student 1 : 30
Student  1  Fail!
Enter marks for student 2 : 20
Student  2  Fail!
Enter marks for student 3 : 40
Student  3  Pass!
Enter marks for student 4 : 40
Student  4  Pass!
Enter marks for student 5 : 40
Student  5  Pass!
```

17. Write a program to print numbers from 1 to 100 and display Even or Odd for each.

```
for i in range(1,101):
  if i % 2 == 0:
    print(i," : Even")
```

```
  else:
    print(i," : Odd")
```

```
1  : Odd
2  : Even
3  : Odd
4  : Even
5  : Odd
6  : Even
7  : Odd
8  : Even
9  : Odd
10  : Even
11  : Odd
12  : Even
13  : Odd
14  : Even
15  : Odd
16  : Even
17  : Odd
18  : Even
19  : Odd
20  : Even
21  : Odd
22  : Even
23  : Odd
24  : Even
25  : Odd
26  : Even
27  : Odd
28  : Even
29  : Odd
30  : Even
31  : Odd
32  : Even
33  : Odd
34  : Even
35  : Odd
36  : Even
37  : Odd
38  : Even
39  : Odd
40  : Even
41  : Odd
42  : Even
43  : Odd
44  : Even
45  : Odd
46  : Even
47  : Odd
48  : Even
49  : Odd
50  : Even
51  : Odd
52  : Even
53  : Odd
54  : Even
55  : Odd
56  : Even
57  : Odd
58  : Even
```

18. Write a program to find how many numbers between 1 and 50 are divisible by 3.

```
count = 0
for i in range(1,51):
  if i % 3 == 0:
    count += 1
print(count)
```

```
16
```

19. Write a program to take marks of 5 students and calculate the average.

```
sum = 0
for i in range(1,6):
  marks = int(input(f"Enter marks for student {i} : "))
  sum += marks
avg = sum/5
print("Average : ", avg)
```

```
Enter marks for student 1 : 10
Enter marks for student 2 : 9
Enter marks for student 3 : 8
```

```
Enter marks for student 4 : 7
Enter marks for student 5 : 6
Average :  8.0
```

20. Write a program to take 5 numbers from the user and count positive and negative numbers

```
pos = 0
neg = 0
for i in range(1,6):
  marks = int(input(f"Enter Integer {i} : "))
  if marks < 0:
    neg -= 1
  else:
    pos += 1
print("Count of Positive integers : ", pos)
print("Count of Negative integers : ", neg)
```

```
Enter Integer 1 : -10
Enter Integer 2 : -11
Enter Integer 3 : -22
```